Unit 3.2 - Derivatives

1 Calculus and Change

Calculus is fundamentally about the study of change. In many real-world situations, we are interested in understanding how a particular quantity changes over time or some other variable. To study change mathematically, we analyze the **rate of change** of a quantity. The rate of change tells us how quickly one variable is changing in relation to another. For example:

- In physics, we might be interested in how the temperature depends on your position in a room.
- In astronomy, we might be interested in how the the force of gravity changes with respect to the distance from a massive object.
- In biology, we examine how populations grow or shrink over time.

2 Derivatives

2.1 Motivation via Secant Lines

One of the central techniques that calculus uses to study change is the *derivative*. This is essentially a generalization of slope that can be calculated at any point on a function's graph. Often times, the derivative is referred to as the *instantaneous slope* since it let's you calculate the slope of a line that is tanget (touches the graph at one point) to the function at a certain point.

To work towards the idea of the derivative, let's consider first a few secant lines (lines which touch a graph at two points). In Figure **??**, you will see several secant lines where the two points get progessively closer together. We can calculate the slope of these lines using the standard formula for the slope of a secant line:

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

where x_1 and x_2 are the x values of the secant points. We can imagine bringing x_2 closer to x_1 (which is held fixed). By doing this, the slope of the secant line will approach a limiting value. This value is the value of the derivative $f'(x_1)$ at x_1.

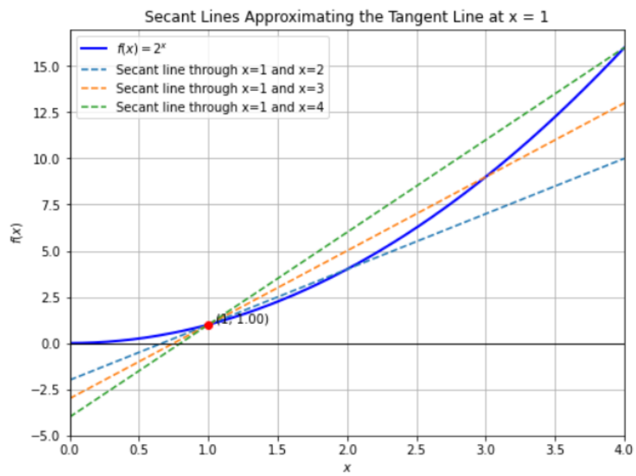

Figure 1: Secant lines getting closer to the tangent line at $x = 1$ for the function $f(x) = x^2$.

2.2 Limits

2.2.1 Intro to Limits

This idea of a limiting value has a formal term and meaning in calculus: the *limit of a function*. Limits are a fundamental concept in calculus that allow us to understand the behavior of functions as they approach specific points (or even infinity). They provide a way to analyze the values that a function approaches, even if it does not actually reach those values.

The notation for a limit is typically expressed as:

$$\lim_{x \to c} f(x) = L$$

The left-hand side is read as, "the limit of $f(x)$ as x goes to c." This means that as x gets arbitrarily close to the value c, the function $f(x)$ approaches the value L. This idea is crucial when dealing with functions that may be undefined at certain points. For example, consider the function:

$$f(x) = \frac{x^2 - 1}{x - 1}$$

This function is undefined at $x = 1$ because the denominator becomes zero. However, we can still evaluate the limit:

$$\lim_{x \to 1} f(x)$$

By simplifying $f(x)$:

$$f(x) = \frac{(x-1)(x+1)}{x-1} = x + 1 \quad \text{for } x \neq 1$$

Thus, as x approaches 1, $f(x)$ approaches 2:

$$\lim_{x \to 1} f(x) = 2$$

2.2.2 One-Sided Limits

Limits can also be approached from one side, leading to one-sided limits. The left-hand limit is expressed as:

$$\lim_{x \to c^-} f(x) = L$$

This indicates that x approaches c from values less than c. The right-hand limit is expressed as:

$$\lim_{x \to c^+} f(x) = L$$

This indicates that x approaches c from values greater than c.

If both one-sided limits exist and are equal, we say that the limit (as a whole) at that point exists. An example of a function where the left and right hand limits disagree would be the function $f(x) = 1/x$, which has the right hand limit

$$\lim_{x \to 0^+} \frac{1}{x} = \infty$$

and the left hand limit

$$\lim_{x \to 0^-} \frac{1}{x} = -\infty$$

2.2.3 Limits at Infinity

Limits can also describe the behavior of functions as x approaches infinity:

$$\lim_{x \to \infty} f(x) = L$$

indicates that as x grows larger and larger, $f(x)$ approaches L. This is particularly useful in understanding horizontal asymptotes. For example,

$$\lim_{x \to \infty} \frac{1}{x} = 0$$

3 Difference Quotient and Derivative

Returning to derivatives, we can now define them using limits via the *difference quotient*. The difference quotient is a formula that provides the slope of a secant line. For a function $f(x)$, the difference quotient is given by:

$$\text{Difference Quotient} = \frac{f(x+h) - f(x)}{h}$$

where:

- x is a point in the domain of the function,
- h is a small increment (the change in x).

This formula calculates the average rate of change of $f(x)$ as we move from x to $x+h$.

To find the derivative $f'(x)$ of the function $f(x)$ at the point x, we consider the limit of the difference quotient as h approaches zero:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

This limit represents the instantaneous rate of change of the function at the point x. If the limit exists, we say that f is *differentiable* at x.

3.1 Example 1: Constant Function $f(x) = a$

For a constant function $f(x) = a$:

$$f'(x) = \lim_{h \to 0} \frac{a - a}{h} = \lim_{h \to 0} \frac{0}{h} = 0$$

Thus, the derivative $f'(x) = 0$.

3.2 Example 2: Linear Function $f(x) = x$

For the linear function $f(x) = x$:

$$f'(x) = \lim_{h \to 0} \frac{(x+h) - x}{h} = \lim_{h \to 0} \frac{h}{h} = \lim_{h \to 0} 1 = 1$$

Thus, the derivative $f'(x) = 1$.

3.3 Example 3: Quadratic Function $f(x) = x^2$

For the quadratic function $f(x) = x^2$:

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \to 0} \frac{2xh + h^2}{h} = \lim_{h \to 0} (2x + h) = 2x$$

Thus, the derivative $f'(x) = 2x$.

3.4 Derivative Notation

There are multiple ways to express the derivative of a function:

$$f'(x) = \frac{df}{dx} = \frac{d}{dx} f(x)$$

The terms df, dx are called *infinitesimals*. You can think of them as “infinitely small” versions of the rise and run (respectively) that you obtain when you take the limit of the difference quotient. The third form write the derivative as an *operator* on $f(x)$, basically some mathematical process that takes an object (in this case, a function) and outputs a new object of the same type (another function which is the derivative of the first).

4 Second Derivatives

In calculus, the second derivative of a function measures how the rate of change itself is changing (the rate of change OF the rate of change). For example, if $x(t)$ is your position as a function of time, then it's first derivative (with respect to time) will be your velocity: $\frac{dx}{dt} = v(t)$. Now, if you take the derivative of velocity (the second derivative of position), you get acceleration: $\frac{d^2x}{dt^2} = \frac{dv}{dt} = a(t)$

The second derivative of a function $f(x)$ is simply the derivative of the first derivative. If $f'(x)$ is the first derivative of the function, the second derivative $f''(x)$ is given by:

$$f''(x) = \lim_{h\to 0} \frac{f'(x+h) - f'(x)}{h}$$

4.1 Example 1: Constant Function $f(x) = a$

For a constant function $f(x) = a$, we already know that the first derivative is zero:

$$f'(x) = 0$$

The second derivative is also zero:

$$f''(x) = \lim_{h\to 0} \frac{f'(x+h) - f'(x)}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0$$

Thus, the second derivative $f''(x) = 0$ for a constant function.

4.2 Example 2: Linear Function $f(x) = x$

For the linear function $f(x) = x$, the first derivative is a constant:

$$f'(x) = 1$$

The second derivative is:

$$f''(x) = \lim_{h\to 0} \frac{f'(x+h) - f'(x)}{h} = \lim_{h\to 0} \frac{1-1}{h} = 0$$

Thus, the second derivative $f''(x) = 0$ for a linear function, which means there is no change in the rate of change (i.e., no acceleration).

4.3 Example 3: Quadratic Function $f(x) = x^2$

For the quadratic function $f(x) = x^2$, we already know that the first derivative is:

$$f'(x) = 2x$$

Now, let's compute the second derivative:

$$f''(x) = \lim_{h\to 0} \frac{f'(x+h) - f'(x)}{h} = \lim_{h\to 0} \frac{2(x+h) - 2x}{h} = \lim_{h\to 0} \frac{2h}{h} = 2$$

Thus, the second derivative $f''(x) = 2$, meaning the rate of change of the slope (or velocity) is constant for a quadratic function.

4.4 Concavity

Since the second derivative deals with second order rates of change, it can describe the curvature of a graph. In particular it descibres the concavity of a function. Concavity describes whether a function bends upwards or downwards at a particular point. A function can be concave up or concave down and the sign of the second derivative can tell us which is the case. For example,

- A function is **concave up** at a point x if its graph is shaped like a cup (bends upwards) near that point. Mathematically, a function is concave up when $f''(x) > 0$. In this case, the slope of the tangent line (the first derivative) is increasing.

- A function is **concave down** at a point x if its graph is shaped like an upside down cup (bends downwards) near that point. This occurs when $f''(x) < 0$, meaning the slope of the tangent line is decreasing.

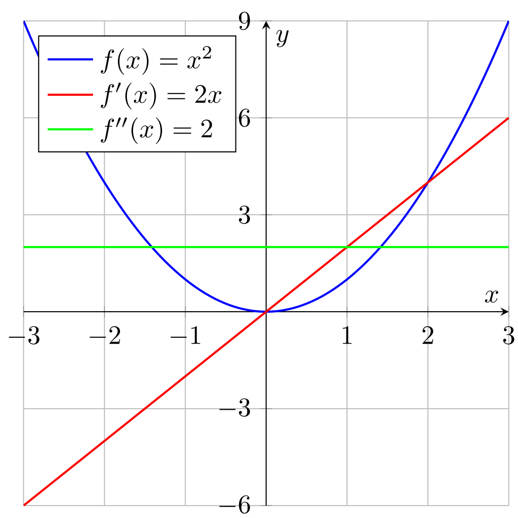

Figure 2: Plot of $f(x) = x^2$ with its first and second derivatives

Figure **??** displays the function $f(x) = x^2$ and both its first and second derivatives. As you can see, the first derivative increases as x increase, causing the function to curve upwards at an increasing rate. Thus, the second derivative is positive and the function is concave up.

A **point of inflection** occurs where the concavity of the function changes from concave up to concave down or vice versa. At a point of inflection, the second derivative zero, i.e., $f''(x) = 0$. However, just because the second derivative is zero at a point doesn't necessarily mean it's an inflection point: it must also involve a change in the sign of the second derivative (consider the obvious example of a constant function which has zero second derivative everywhere, but is not concave).

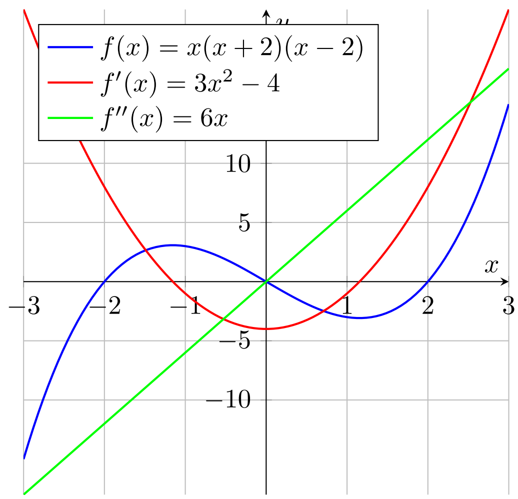

Figure 3: Plot of $f(x) = x(x+2)(x-2)$ with its first and second derivatives

In Figure **??**, you can see a plot of $f(x) = x(x+2)(x-2)$ and its derivatives. The function is concave down for negative x and concave up for positive x. At $x = 0$ there is an inflection point (the second derivative plot equals zero).

5 Numerical Derivatives

In calculus, the derivative of a function is defined as a limit. However, on computers, we are constrained by *finite precision* due to the limitations of floating-point arithmetic. This means that we cannot take true limits, and as a result, all derivatives calculated on a computer are *approximations*.

5.1 Challenges of Finite Precision and Noise

When calculating numerical derivatives, we are approximating the rate of change of a function by using a small but finite *step size* (the computational equivalent of the h in the difference quotient). The choice of step size is crucial. If the step size is too large, we lose accuracy because we deviate from the true derivative definition (we end up with a secant line instead of tanget line). If the step size is too small, floating-point errors can dominate, making the result less reliable. Furthermore, data noise (either from measurement error or inherent instability) also introduces issues when calculating numerical derivatives. Small fluctuations in the function's value can significantly affect the accuracy of the derivative estimate, especially when the step size is small.

5.2 Finite Difference Methods

There are two cases where derivatives must be computationally evaluated:

- You may have a representation of your function written in code that you can simply pass an argument to and get back an output. In this case, you have a great degree of freedom in choosing the step size so long as you're wary of the above issues.
- Alternatively, you may just be given a list $\{(x_1, t_1), \cdots, (x_n, t_n)\}$ of fixed data points. In this case, measurements were only taken at these particular times and you can't use a step szie smaller than the time gap between them without using more advanced techniques like regression or interpolation.

In either case, there are several common approaches to approximating the derivative of a function numerically, based on *finite difference methods*. These methods use the values of the function at different points to approximate the slope.

5.2.1 Forward Difference

The forward difference method approximates the derivative at a point x using the value of the function at $x + h$, where h is a small step size:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

As you can see, this is identical to the difference quotient. This method is simple to compute, but it tends to introduce more error than more sophisticated methods, as it only uses information from one side of the point x.

In cases where you only have data points, the formula becomes:

$$y_i' \approx \frac{y_{i+1} - y_i}{t_{i+1} - t_i}$$

for all $i = 1, \cdots, n-1$. Notice that since this formula requires two points, we end up with one less point than we started with. Often times, measurements are taken at regular intervals, meaning that $t_{i+1} - t_i = \Delta t$ for all i, simplfying the formula slightly.

5.2.2 Backward Difference

The backward difference method uses the value of the function at $x - h$ to approximate the derivative:

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

This method is similar to the forward difference but approaches the derivative from the left side of x.

In cases where you only have data points, the formula becomes:

$$y_i' \approx \frac{y_i - y_{i-1}}{t_i - t_{i-1}}$$

Again, one point is lost.

5.2.3 Central Difference

The central difference method works by averaging the forward and backward differences. It uses values of the function on both sides of x:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

This method is generally more accurate than the forward or backward difference methods because it accounts for information on both sides of the point x.

In cases where you only have data points, the formula becomes:

$$y_i' \approx \frac{y_{i+1} - y_{i-1}}{t_{i+1} - t_{i-1}}$$

In this case, two points are lost.

5.2.4 Numerical Second Derivatives

Similarly, second derivatives can also be approximated using finite differences. Let us start with the data points case this time. If we apply the forward difference method to the first derivative, we get:

$$y_i'' \approx \frac{y_{i+1}' - y_i'}{\Delta t} =$$

Now, if we replace the y_i', y_{i+1}' with their backward difference formulae and simplify:

$$y_i'' \approx \frac{y_{i+1}' - y_i'}{\Delta t} = \frac{\frac{y_{i+1} - y_i}{\Delta t} - \frac{y_i - y_{i-1}}{\Delta t}}{\Delta t} = \frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta t^2}$$

Similarly, the case with a function is given by:

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

5.3 Noisy Derivatives and Rolling Averages

When computing numerical derivatives, the presence of noise in the data can lead to inaccurate and unstable results. Noise can stem from various sources, such as measurement errors or finite precision in computer calculations. Particularly when the step size is small, the noise can overwhelm the derivative estimate.

A common approach to addressing this issue is to apply a *rolling average* to smooth the data before calculating derivatives. This technique helps to reduce the influence of noisy data points and provides a more stable derivative approximation.

5.3.1 Rolling Average Formula

The rolling average smooths the function values over a small window of nearby points. For a dataset $\{(t_i, y_i)\}$, the rolling average for a window of size $2N+1$ centered at t_i is:

$$\text{Rolling Average}(y_i) = \frac{1}{2N+1} \sum_{j=-N}^{N} y_{i+j}$$

This essentially averages the point in question with the N values in front of it and behind it. You can think of the window as having a radius of N points. This smooths out fluctuations caused by noise, making the subsequent numerical derivative more reliable. Once the rolling average is applied, the derivative can be computed using finite difference methods.

5.3.2 Trade-offs in Smoothing

While applying a rolling average helps reduce noise, there is a trade-off. If the window size N is too large, the smoothing can mask important local variations in the data, leading to an inaccurate representation of the derivative which is overly flat and smooth. Therefore, it is crucial to select an appropriate window size to balance noise reduction and accuracy. The choice will be problem dependent and involve trail and error.